

# Orchestrating Reasoning and Reaction: An Asynchronous Hierarchical Framework for LLM-driven Traffic Signal Control

*Fansheng Sun , Jiyu Wang , Zhidan Liu**

*The Hong Kong University of Science and Technology (Guangzhou)*

** Corresponding author: zhidanliu@hkust-gz.edu.cn*

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## 1 Background & Motivation

**TSC dynamically allocates green-light durations across urban intersections. Its goal is to reduce waiting time, improve traffic throughput, and prevent congestion from propagating across the road network..**

### MARL-based Methods

**Fast adaptive control through local interaction and neighborhood communication.**
**However, limited observation leads to myopic coordination and weak global awareness.**

### LLM-driven Agents

**Strong semantic reasoning and zero-shot strategic planning for network-wide TSC.**
**However, direct LLM control suffers from latency, high context cost, and topological hallucination.**

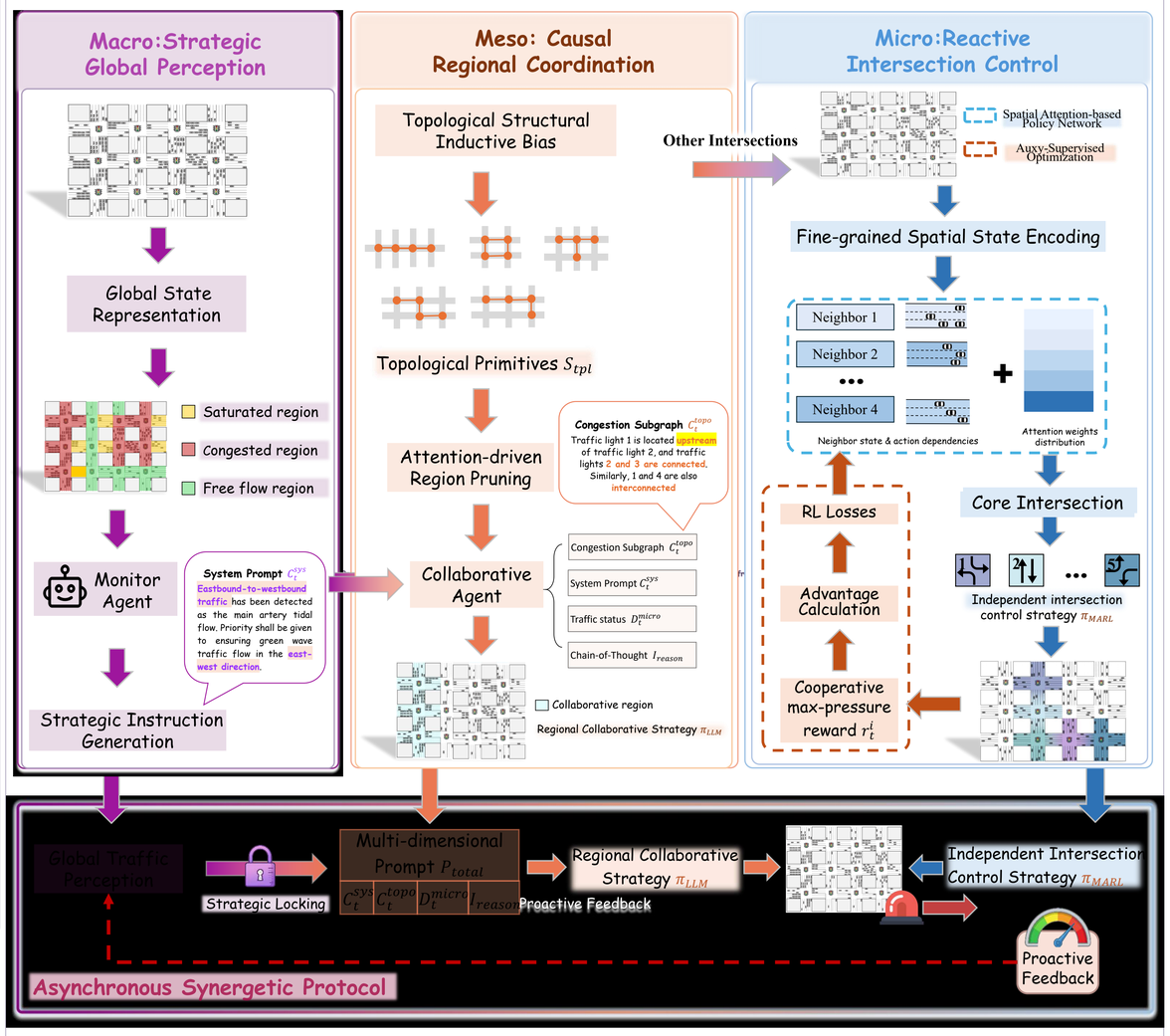
### Our Gogal

**Bridge strategic LLM reasoning and reactive MARL control across different time scales.**
**Achieve topology-grounded network-wide TSC through hierarchical asynchronous synergy.**

## Key Challenges

- **C1** Computational burden vs. real-time response
- **C2** Spatial-topological mismatch
- **C3** Asynchronous coordination instability

## 2 Our Proposal: Astra




1. **Macro:** Global bottleneck detection via low-frequency LLM reasoning.
2. **Meso:** Topology-grounded causal coordination over congestion subgraphs.
3. **Micro:** High-frequency MARL control with spatial attention.
4. **Protocol:** Asynchronous locking and feedback for stable re-planning.

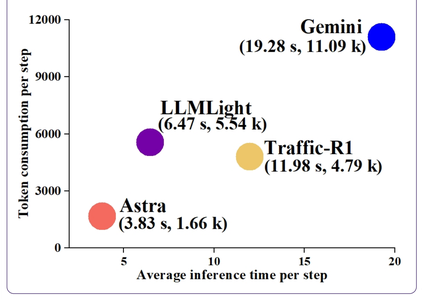
## 3 Experimental Results

### A. Overall Performance

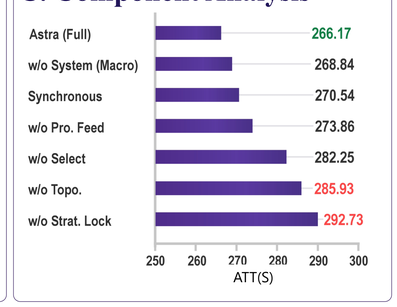
| Model | Jinan | | Hangzhou | |
|---|---|---|---|---|
| | ATT | AWT | ATT | AWT |
| FixedTime | 358.49 | 297.23 | 470.92 | 297.5 |
| Advanced CoLight | 283.94 | 55.16 | 336.12 | 50.44 |
| LLMLight | 275.74 | 58.42 | 306.53 | 39.27 |
| Traffic-R1 | 264.48 | 47.93 | 308.46 | 43.94 |
| Dec-POMDP | 285.36 | 59.24 | 310.14 | 43.32 |
| **Astra (Gemini-3-Flash)** | **253.28** | **45.03** | **301.95** | **21.66** |

Astra consistently achieves **SOTA** performance across 5 real-world datasets.

### B. Efficiency: Tokens vs. Time




### C. Component Analysis



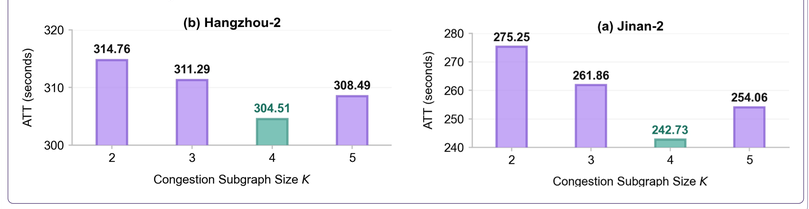

### D. Large-scale Evaluation (Lower is Beter)

196 INTERSECTIONS

| Method | New York-1 (196 intersections) | | New York-2 (196 intersections) | |
|---|---|---|---|---|
| | ATT (s) | AWT (s) | ATT (s) | AWT (s) |
| Advanced-CoLight | 1180.56 | 141.46 | 1467.89 | 261.37 |
| Traffic-R1 | 932.74 | 77.24 | 1194.69 | 197.04 |
| CoLLMLight | 971.88 | 93.38 | 1235.37 | 198.49 |
| **Astra (Qwen3-30B)** | **923.18** | **70.36** | **1140.82** | **154.84** |

Astra sustains **SOTA** performance on city-scale networks.

### E. Sensitivity




## 4 Key Contributions

**Macro–Meso–Micro Hierarchy**
A principled three-level hierarchy that integrates strategic reasoning and reactive control.

**Topology-grounded LLM Coordination**
Semantic causal primitives align LLM reasoning with the physical road structure.l.

**Asynchronous Synergetic Protocol**
Strategic locking and proactive feedback synchronize planning and reaction effectively.

**Extensive Real-world Validation**
Evaluated on 5 real-world datasets and 2 large-scale networks (196 intersections), achieving consistent SOTA results.

**Astra is the SOTA across 5 real-world datasets and 2 city-scale networks.**